Ashish S

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Summary

Data Engineer and ML practitioner with **2+ years** building production-grade data pipelines, AI agent infrastructure, and ML systems at scale. Delivered **8x pipeline throughput improvement** and led technical onboarding of **Meta** onto an AI procurement platform. Experienced across **Spark, BigQuery, Snowflake, GCP, and AWS** with hands-on ML deployment spanning vendor recommendation, demand forecasting, and NLP - backed by a **1st place finish** in a national data science competition among 500+ participants.

Experience

Data Engineer Mar 2025 - Present
Aerchain - AI Procurement Platform *India*

- **AI Agent Data Infrastructure:** Engineered automated **Spark + BigQuery** data pipelines serving 3 production AI agents (Negotiation Agent, Procurement Agent, and Aera scenario simulation), processing procurement transactions across **1M+ vendor records** with sub-minute refresh latency.
- **Pipeline Optimization:** Redesigned batch ingestion architecture on **Snowflake**, reducing end-to-end pipeline runtime from **8 hours to 2 hours** (4x improvement) through parallelization, partition pruning, and incremental load strategies.
- **Meta Client Onboarding:** Served as **technical lead** on a 3-person team to onboard Meta, owning API integration, data migration, pipeline setup, and demo environment configuration; directly contributed to closing a strategic enterprise account.
- **Smart Vendor Recommendation:** Built a **Random Forest** vendor recommendation engine on **1M+ historical purchase records**, surfacing ranked supplier shortlists based on price, quality, and category signals - deployed to production.
- **ML Scenario Builder:** Developed a supply chain simulation module for Aera using historical procurement data to generate realistic stress-test scenarios, enabling clients to model disruption risk across supplier networks.

Data Analyst Dec 2023 - Jan 2025
Deloitte USI *India*

- **Telecom Data Integration:** Built and maintained **Oracle Integration Cloud** pipelines processing **5M+ daily telecom transactions** at **99.8% availability**, ensuring reliable data delivery to downstream billing systems.
- **Query Optimization:** Tuned **15+ high-volume SQL/PL-SQL** queries via indexing and execution plan analysis, reducing runtime from **55s to 15s** and cutting compute costs by **50%**.
- **Executive Dashboards:** Delivered **Power BI and Tableau** dashboards on **10GB+ enterprise datasets**, enabling data-driven decisions across **8 cross-functional teams**.

R&D Intern - NLP Feb 2023 - Jul 2023
ExcelSoft Technologies *India*

- **Multilingual NLP:** Processed **500K+ text samples** across 3 regional Indian languages using custom Python tokenizers; improved model **F1-score from 68% to 88%** via corpus cleaning and code-switching normalization.

Projects

Netflix Pivot & Recommendation Engine | *Genetic Algorithm, Random Forest, Python* Live Demo

- Solved a strategic recommendation problem on whether Netflix should pivot content strategy and how to optimize content recommendations. Applied **Genetic Algorithm** for feature selection and **Random Forest** ensemble for preference modeling - ranked **1st among 250 participants** across competing institutions.

Precision Agriculture Decision Engine | *Python, Scikit-learn, IoT, REST API, JavaScript* Live Demo

- Built an IoT analytics platform ingesting **4 real-time sensor streams** (soil moisture, temperature, humidity, pH); trained **Random Forest** classifier for adaptive irrigation scheduling, reducing simulated water usage by **30%** vs rule-based baseline.

Technical Skills

Languages: Python (primary), SQL, PL/SQL, R, Java
ML & Data Science: Scikit-learn, XGBoost, Random Forest, Genetic Algorithms, TensorFlow, PyTorch, HuggingFace, spaCy
Data Engineering: Apache Spark, dbt, ETL Pipelines, Oracle Integration Cloud, FastAPI, Docker, REST APIs
Cloud & Databases: GCP (BigQuery), Snowflake, AWS (S3, EC2), PostgreSQL, MySQL, Oracle DB, Git
Analytics & BI: Power BI, Tableau, Pandas, NumPy, A/B Testing, Statistical Analysis

Education

The National Institute of Engineering Mysuru, India
B.E. in Electronics & Communication Engineering ***- CGPA: 8.85/10*** *Aug 2019 - Oct 2023*

Awards & Achievements

Top 100 among **35,000+ participants** in Google's Prompt Wars; secured **1st Place** in the IMI Data Science Competition among **500 participants**; **Semi-Finalist** in Amazon AI for Bharat.

Recipient of the **IEI Engineering Award**, presented by the **Chief Minister of Karnataka** for an outstanding academic project.

Elected **President, Onyx Entrepreneurship Club** and **IEEE Student Activity Coordinator** - drove technical and innovation initiatives institute-wide.

Achieved **KCET Rank 2282** among **500K+ candidates** in the Karnataka state engineering entrance examination.